# ASK THE CATALOG

Group 5 • MSDS 630 NLP Hackathon • October 9, 2026

**OFFLINE-FIRST PDF RETRIEVAL + GROUNDED EXTRACTIVE ANSWERS**

Provisional source set — not instructor-approved catalog PDFs. Six project-created snapshots from public official UC webpages; the 2026–27 catalog host returned HTTP 403 and handbook access was restricted. Professor approval or a rerun on original catalog/handbook PDFs is still needed.

## DATA AND METHODS

Sources: executive student resources, registrar, satisfactory academic progress, student complaints, accommodations, and admission requirements. Captured October 9, 2026. 6 PDFs → 39 extracted pages → 74 page-bound chunks (350 words with 50-word overlap; PyMuPDF). The UI exposes one-based PDF page citations and the original source texts.

Retrieval: baseline TF-IDF word 1–2 grams + cosine; final hybrid 70% word + 30% character 3–5-gram TF-IDF. Top five chunks feed source-sentence excerpts; citation IDs identify PDFs/pages. Unsupported and personal/future/live questions trigger refusal. Optional local embeddings and local Ollama generation are available but not part of the reported experiment.

## MEASURED DEVELOPMENT RESULTS | 25 ANSWERABLE + 5 UNANSWERABLE

| METRIC | BASELINE | CAL. TF-IDF | HYBRID |
|---|---|---|---|
| Gold-passage Recall@1 | 18/25 | 18/25 | **16/25** |
| Gold-passage Recall@5 | 23/25 | 23/25 | **24/25** |
| Correct refusals | 3/5 | 5/5 | **5/5** |
| False refusals | 2/25 | 0/25 | **0/25** |
| Mean reciprocal rank | 0.790 | 0.790 | **0.755** |

Passage-level Recall@5 requires the exact gold supporting phrase inside a correctly cited top-five chunk. Separate 8-question calibration selected thresholds 0.095 / 0.118. The 30 were used during later iteration: these are not blind held-out results. Seventeen tests passed. Displayed answer excerpts contain the gold quote in 24/25 answerable cases, which is not semantic answer accuracy.

## OBSERVED ERRORS, ETHICS, AND NEXT STEPS

Q10: INTR599 supporting chunk was below rank 5. Q01/Q09: supporting passages were rank 4 and initially omitted when only three excerpts were shown. Q13/Q15: short sentence selection omitted Parchment or the second degree deadline; expanded context fixed visibility. Hybrid improves Recall@5 but lowers Recall@1 (64% vs 72%). Citation presence is not proof of truthful interpretation; semantic correctness/faithfulness still require independent human review. Next: professor-approved corpus, fresh blinded QA set, OCR, table checking, and reviewer-scored answer/citation quality.

## FIVE CONTRIBUTIONS AND DISCLOSURE

1 Venkat: integration/README/tests; 2 Data: PDFs/provenance/chunking; 3 Modeling: TF-IDF/hybrid/refusal; 4 Evaluation: 30 QA, metrics/errors; 5 UI: Streamlit, demo, slides. ChatGPT/Codex assisted with the implementation, QA drafting, analysis, report and slides. Student verification, instructor document approval, and real-device rehearsal remain the group's responsibility.